Sentinel

Risk Intelligence for FINMA-regulated banks

Explainable AI for cross-jurisdictional compliance review.

SwissHacks 2026 · Team Sentinel

The problem

Compliance officers at AMINA, Julius Baer, and similar banks review **hundreds of flagged cases per day** — voice transfers, suspicious trades, on-chain transactions.

Each case needs:

- A **risk score** they can defend in writing
- An explanation that survives a **regulator audit**
- Reasoning that works across **4 jurisdictions** (FINMA, MiCA, SFC, FSRA)
- A **decision trail** that holds up in court

Current tools give them a black-box score. Officers either rubber-stamp or override blindly. **Liability falls on them.**

What we built

A single dashboard where the compliance officer sees:

- **The score** — XGBoost, 0-100
- **Why** — SHAP feature contributions, ranked
- **What would change it** — DiCE counterfactual scenarios
- **Under whose rules** — live toggle between FINMA / MiCA / SFC / FSRA
- **What goes to AI** — privacy split-view (FINMA Circular 2024/3)
- **Their decision** — immutably logged with rationale if overriding AI

End-to-end. Streaming. Auditable.

Live demo: Marc Weber case

CRITICAL · 100/100

> **Voice call** requesting CHF 8.7M to unknown wallet
> Sunday 3:14am · Destination: RU
> Transcript: *"urgent, my partner is waiting, don't tell anyone..."*

In the next 90 seconds:

1. AI assessment streams in word-by-word
2. Top 5 risk factors light up
3. Counterfactual: *"if destination weren't Russia..."*
4. Toggle to **FSRA** — score adjusted
5. Decision: **BLOCK** — logged with rationale

→ Switching to demo

Four things we did differently

	Most teams	Sentinel
Explainability	SHAP only	SHAP **+ DiCE counterfactuals**
LLM data	Raw client info	**Anonymized pseudonyms** + bucketed amounts
Jurisdictions	Hardcoded	**YAML rule packs**, live toggle
UX	Request → wait → response	**Server-Sent Events** streaming

These aren't features bolted on. They're architectural decisions because AMINA actually operates under four regulators.

Architecture

```
Frontend (Next.js 15)                    Backend (FastAPI)
─────────────────────                    ─────────────────
Case Queue                ──REST──▶       Risk Engine
                                              │
Detail Panel              ◀──SSE──         XGBoost + SHAP
   ├ Streaming AI                             │
   ├ SHAP viewer                         DiCE Counterfactuals
   ├ Counterfactuals                          │
   ├ Privacy split                       Jurisdiction Engine
   ├ Jurisdiction toggle                    (4 YAML packs)
   └ Decision bar                             │
                                          Anonymizer
                                              │
                                       Claude API · SSE stream
                                              │
                                       Audit Log (immutable)
```

19 API endpoints · 4 jurisdiction rule packs · persistent DB · mock-mode fallback

Privacy by design — what FINMA cares about

Before any AI call, the anonymizer transforms case data:

What stays local	What goes to AI
`client_name: "Marc Weber"`	`client_pseudonym: "CLIENT_AAF7"`
`amount_chf: 8,700,000`	`amount_band: "CHF 5M–10M"`
`destination_wallet: 0x3a1b...`	`destination_wallet: "0xUN****9012"`
`voice_sample_id: vs_001_...`	redacted
`transcript_excerpt: ...`	redacted

The model reasons about **patterns**, not personal data.

Compliance officer audits this **before** signing off.

Cross-jurisdictional reasoning

Same case. Different rules.

Jurisdiction	Adjusted score	Recommended action
CH · FINMA	100	**BLOCK**
EU · MiCA	100	**BLOCK**
HK · SFC	92	ESCALATE
AE · FSRA	100	**BLOCK**

Each rule pack is **YAML**. Compliance team edits without writing code.

This is the AMINA cross-border challenge — directly addressed.

What's next

On the hackathon weekend — features the team will add:

- **Voice biometric layer** for AMINA challenge (deepfake detection)
- **Julius Baer skin** — investment recommendation walkthrough
- **Ripple skin** — XRPL transaction with RLUSD escrow flow
- **Real-time alerts** — WebSocket notifications

Already built and waiting — backend supports all three case types via the same engine. Adding skins is hours, not days.

Thank you

Repo: github.com/SteveDok22/swisshacks-2026

Live demo: localhost:3000

Questions?